



معسكر الورقة الثانية للأطفال — اليوم الأول



Day 1 Quiz – Second Paper Camp



Answer Key Edition

Total Marks: 45 | Time: 30 min

QUIZ TYPE

Interactive Matching

TOTAL MARKS

45 Marks

DURATION

30 Minutes

Matching Questions sets

Total: 45 marks

1 Theme: Acute Upper Airway Infections & Obstruction (5 marks)

- A. Viral Croup
- B. Acute Epiglottitis
- C. Acute Bronchiolitis
- D. Foreign Body Aspiration
- E. Spasmodic Croup

From the list above, select one choice that is most suitable for the following options:

- 1 A 6-month-old infant presents during winter with a first episode of expiratory wheezing, tachypnea, and chest retractions, following a 3-day history of coryza and low-grade fever.
- 2 Anteroposterior neck radiography of a 2-year-old child presenting with a barking cough and inspiratory stridor demonstrates subglottic narrowing known as the "steeple sign".
- 3 A 3-year-old child presents with a sudden onset of barking cough at night without a preceding viral prodrome or fever, which completely resolves within a few hours.
- 4 A 4-year-old unimmunized child presents with a toxic appearance, high fever, severe respiratory distress, drooling of saliva, and insistence on sitting forward in a tripod position.
- 5 A 14-month-old healthy toddler experiences a sudden episode of choking and coughing while playing, followed by a fixed localized wheeze and decreased breath sounds on one side.

2 Theme: Diagnostic & Management Safety Maneuvers (5 marks)

- A. Expiratory chest radiograph
- B. Direct oral examination with a tongue depressor
- C. Spirometry showing reversibility of airway obstruction
- D. Nebulized Epinephrine followed by a 2-hour observation period
- E. Routine administration of sedatives

From the list above, select one choice that is most suitable for the following options:

- 1** A diagnostic evaluation modality used to confirm the diagnosis of Bronchial Asthma by documenting the reversible nature of the airway obstruction.
- 2** A strictly prohibited procedure in a child suspected of acute epiglottitis due to the critical risk of precipitating fatal laryngospasm before securing the airway.
- 3** The preferred radiological technique used to visualize obstructive emphysema or localized air trapping caused by a radiolucent foreign body.
- 4** A fast-acting therapeutic intervention used to decrease subglottic edema in severe croup, requiring monitoring due to the risk of rebound symptom recurrence.
- 5** A management intervention that is strictly avoided and contraindicated in the standard supportive care of acute bronchiolitis.

3 Theme: Differential Diagnosis & Pneumonia Mimics (5 marks)

- A.** Diabetic Ketoacidosis (DKA)
- B.** Congenital Lung Anomalies (e.g., CCAM/Sequestration)
- C.** Bronchial Asthma
- D.** Heart Failure
- E.** Löffler Syndrome

From the list above, select one choice that is most suitable for the following options:

- 1** A metabolic condition that mimics respiratory distress by causing deep, rapid respiration (Kussmaul breathing), while the chest examination remains entirely clear.
- 2** An underlying structural etiology that must be highly suspected in a child presenting with recurrent episodes of pneumonia restricted to the exact same lung site.
- 3** Characterized by chronic airway inflammation, bronchial hyperresponsiveness, and recurrent acute episodes triggered by exposure to allergens.
- 4** A non-respiratory systemic condition presenting with tachypnea, cardiomegaly, a gallop rhythm, and a tender enlarged liver, which can be mistaken for childhood pneumonia.
- 5** Characterized by transient pulmonary eosinophilic infiltrates accompanied by mild respiratory symptoms, often secondary to parasites or drugs.

4 Theme: Pediatric Pneumonia Pathogens (5 marks)

- A.** Staphylococcus aureus
- B.** Mycoplasma pneumoniae
- C.** Chlamydia trachomatis
- D.** Viral Pneumonia (e.g., RSV/Adenovirus)
- E.** Streptococcus pneumoniae

From the list above, select one choice that is most suitable for the following options:

- 1** A school-aged child presenting with an insidious onset of atypical pneumonia, where treatment with Macrolides is the preferred therapeutic choice.
- 2** A chest radiograph of a severely ill infant reveals pathogen-specific lung complications such as pneumatoceles, empyema, or pyopneumothorax.
- 3** Characterized by a laboratory profile showing a normal to mildly elevated white blood cell count with a distinct lymphocyte predominance.
- 4** The most common cause of bacterial pneumonia, characterized by a high white blood cell count with polymorphonuclear (PMN) predominance and markedly elevated ESR and CRP.
- 5** An infant presenting with subacute pneumonia, a characteristic staccato cough, and no fever, typically acquired during delivery through an infected birth canal.

5 Theme: Pneumonia Classifications & Hospitalization (5 marks)

- A.** Community-Acquired Pneumonia (CAP)
- B.** Hospital-Acquired Pneumonia (HAP)
- C.** Ventilator-Associated Pneumonia (VAP)
- D.** Age < 6 months
- E.** Fast breathing (Tachypnea)

From the list above, select one choice that is most suitable for the following options:

- 1** A type of lower respiratory infection that presents clinically in a child more than 48 hours after hospital admission.
- 2** A specific category of nosocomial pneumonia that characteristically develops more than 48 hours following endotracheal intubation.
- 3** An absolute age-based indication for hospitalizing a child diagnosed with pneumonia, regardless of the initial mild clinical scoring.
- 4** Defined as signs and symptoms of pneumonia developing in a previously healthy child due to an infection acquired outside a hospital facility.
- 5** The primary clinical sign used by the WHO classification to identify pneumonia in a child presenting with a cough or difficult breathing.

6 Theme: Bronchial Asthma Triggers & Clinical Indicators (5 marks)

- A.** Insidious onset of acute asthma episodes
- B.** Sudden onset of acute asthma episodes
- C.** Pulsus paradoxus
- D.** Palpable liver and spleen margins
- E.** Day-to-day and/or morning-to-evening variation of PEF or FEV1 $\geq$ 20%

From the list above, select one choice that is most suitable for the following options:

- 1** The typical clinical onset pattern observed when an acute asthma episode is triggered following viral infections.
- 2** The typical clinical onset pattern observed when an acute asthma episode is caused by exposure to irritants like cold air or noxious fumes.
- 3** A cardiovascular clinical sign characterized by an exaggerated drop in systolic blood pressure during inspiration, accompanying severe asthma attacks.
- 4** A physical examination finding in severe asthma that occurs entirely as a result of chest hyperinflation displacing the diaphragm downward.
- 5** A diagnostic parameter obtained on lung function tests that indicates significant diurnal variation and airway reversibility in a child with asthma.

7 Theme: Bronchial Asthma Lab Profiles & Comorbidities (5 marks)

- A.** Early-stage acute asthma exacerbation
- B.** Late-stage impending respiratory failure
- C.** Gastroesophageal Reflux Disease (GERD)
- D.** Eosinophilia
- E.** Silent chest

From the list above, select one choice that is most suitable for the following options:

- 1** A chronic comorbid condition commonly associated with bronchial asthma that induces recurrent micro-aspiration, making the airway disease difficult to control.
- 2** A blood laboratory finding that serves as a useful indicator of the underlying allergic or atopic profile in a child with bronchial asthma.
- 3** An arterial blood gas (ABG) profile during an initial asthma attack showing hypoxia, low PCO₂, and respiratory alkalosis due to hyperventilation.
- 4** An arterial blood gas (ABG) profile showing a rising or normal PCO₂ accompanied by respiratory acidosis, indicating severe respiratory muscle fatigue.
- 5** A critical auscultatory finding during a severe asthma attack where air entry is so poor that wheezing disappears, indicating imminent respiratory arrest.

8 Theme: Acute Bronchiolitis Pathophysiology & Complications (5 marks)

- A.** Apneic spells
- B.** Complete plugging of airways by mucus and sloughed cells
- C.** Humidified oxygen
- D.** Ribavirin
- E.** Respiratory Syncytial Virus (RSV)

From the list above, select one choice that is most suitable for the following options:

- 1** A critical life-threatening complication that must be closely anticipated in young infants below 2 months of age presenting with acute bronchiolitis.
- 2** The primary supportive therapeutic agent routinely administered in acute bronchiolitis to manage documented hypoxemia.
- 3** A specific antiviral agent that may be considered for the treatment of acute bronchiolitis in highly selected high-risk or congenital cases.
- 4** The pathological mechanism in the bronchioles that directly results in hyperinflation, air trapping, and ventilation-perfusion mismatch.
- 5** The most common underlying viral etiology responsible for the vast majority of cases of acute bronchiolitis in infants.

9 Theme: WHO Tachypnea Definitions & Critical Airway Rules (5 marks)

- A.** Respiratory rate ≥ 50 breaths/minute
- B.** Respiratory rate ≥ 40 breaths/minute
- C.** Avoid placing the child in a supine position
- D.** Avoid phlebotomy or IV line placement before airway control
- E.** Respiratory rate ≥ 60 breaths/minute

From the list above, select one choice that is most suitable for the following options:

- 1** The strict WHO threshold value used to define fast breathing (tachypnea) in an infant aged 2 months up to 12 months.
- 2** The strict WHO threshold value used to define fast breathing (tachypnea) in a child aged 12 months up to 5 years.
- 3** An immediate positioning safety protocol in epiglottitis required because gravity can cause the swollen epiglottis to occlude the airway completely.
- 4** A critical medical safety restriction in epiglottitis aimed at preventing any painful or stressful interventions that could trigger fatal laryngospasm.
- 5** The strict WHO threshold value used to define fast breathing (tachypnea) in a young infant less than 2 months of age.

Model Answers

Theme 1: Acute Upper Airway Infections & Obstruction

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
C	A	E	B	D

Theme 2: Diagnostic & Management Safety Maneuvers

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
C	B	A	D	E

Theme 3: Differential Diagnosis & Pneumonia Mimics

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
A	B	C	D	E

Theme 4: Pediatric Pneumonia Pathogens

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
B	A	D	E	C

Theme 5: Pneumonia Classifications & Hospitalization

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
B	C	D	A	E

Theme 6: Bronchial Asthma Triggers & Clinical Indicators

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
A	B	C	D	E

Theme 7: Bronchial Asthma Lab Profiles & Comorbidities

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
C	D	A	B	E

Theme 8: Acute Bronchiolitis Pathophysiology & Complications

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
A	C	D	B	E

Theme 9: WHO Tachypnea Definitions & Critical Airway Rules

No. 1 (1 mark)	No. 2 (1 mark)	No. 3 (1 mark)	No. 4 (1 mark)	No. 5 (1 mark)
A	B	C	D	E